Greedy Positive-Cone Reduction: from Three Baselines to a Standard Angular-Defect Greedy

Experimental summary

July 17, 2026

Abstract

We build a reduced *positive cone* $K = \mathrm{span}_+\{w_1, \dots, w_R\}$ approximating a set of non-negative snapshots, and compare three greedy constructions — Cone-Projected Greedy (CPG), Angular-Defect Greedy (ADG), modified CPG (mCPG) — on a low-dimensional *lambda* dataset and a high-dimensional *physics* contact dataset. The note is organised as the *evolution* of the method: a fair baseline comparison exposes two weaknesses of the raw greedy cones (poor worst-case relative accuracy when snapshot magnitudes vary, and severe ill-conditioning from near-duplicate generators); we then add one ingredient to ADG — *per-snapshot normalisation* — and call the result the **standard ADG**. It addresses the first weakness and not the second. At equal cone size the standard ADG is 15–28× more accurate than CPG/mCPG on the wide-magnitude physics data (magnitude ratio 604); on the well-scaled lambda data it selects the same generators as CPG, inherits the same $\kappa_2 \sim 10^{10}$, and mCPG dominates both. Conditioning is therefore mCPG's contribution, not ADG's. We then re-test the story on four independent contact-force datasets (membrane obstacle, and three Hertz-contact cases), in their dual W-inner product and with held-out parameters. They confirm the magnitude-ratio thesis on data it was not developed on, and force two refinements: normalisation is *harmful*, not merely inert, on well-scaled data — including on lambda, where it was first called inert; and CPG and ADG are *not* the same algorithm, despite selecting identically on every dataset the method was developed on.

1 Setup

Algorithms. All three methods grow a positive cone one generator at a time (ADG admits every snapshot tied at the maximal angle, so a round may add more than one) and project each snapshot x onto the current cone by non-negative least squares (NNLS), $\Pi_K(x) = \arg\min_{c \geq 0} \|Ac - x\|_2$ with A the generators.

- CPG adds the snapshot of largest *absolute* residual $\|x - \Pi_K(x)\|_2$.
- ADG adds the snapshot of largest *angle* $\theta_K(x) = \arcsin\big(\|x - \Pi_K(x)\|_2 / \|x\|_2\big)$ to the cone — a *scale-invariant* novelty measure, so its selection is intrinsically insensitive to snapshot magnitude.
- mCPG selects like CPG but stores a shifted, re-normalised generator $\nu = (x - \Pi_K(x))/\|x - \Pi_K(x)\|_2$, carrying only the direction new to the cone.

Metric. We report the *per-snapshot relative residual* $\rho(x) = \|x - \Pi_K(x)\|_2 / \|x\|_2$, its worst case $\rho_{\max}$ and mean $\bar{\rho}$ over the dataset, evaluated on the *original, physical-scale* snapshots for every method, plus the basis condition number $\kappa_2 = \sigma_{\max}/\sigma_{\min}$.

Datasets. Table 1 lists the six datasets. The magnitude ratio $\max_x \|x\|_2 / \min_x \|x\|_2$ is the single quantity that predicts which method wins.

Table 1: Datasets. The magnitude ratio (over nonzero snapshots) governs the qualitative differences below. The four contact-force sets (Sec. 5) are measured in their dual W-norm. Every set here except physics is well-scaled (ratio ≤ 4), which by the thesis of this note predicts mCPG-like behaviour and a normalisation that costs more than it returns — borne out on lambda itself (Sec. 3) and on all four contact sets (Sec. 5). Hertz (a) is the only sample at an *intermediate* ratio, and Sec. 5.5 uses it to place a lower bound on where normalisation starts to pay.

Dataset	Snapshots × dim	min $\|x\|$	max $\|x\|$	Magnitude ratio
Lambda	50×57	0.283	0.319	1.13
Physics	96×7676	9.04×10^6	5.46×10^9	604
Hertz	81×47	1.036	1.204	1.16
Membrane	125×885	1.043	1.353	1.30
Hertz (a)	31×51	0.123	0.482	3.93
Hertz (b)	23×51	0.101	0.109	1.09

2 Baseline: the three methods as-is

Run at a common relative tolerance $\varepsilon = 10^{-2}$ (each method stops at its own cone size), the raw greedy cones already reveal the two problems that motivate everything else (Table 2).

Table 2: Baseline at $\varepsilon = 10^{-2}$ (whole dataset). Physics: ADG's scale-invariant selection halves the worst-case error of CPG/mCPG. Lambda: mCPG is best and, crucially, CPG/ADG are numerically singular ($\kappa_2 \sim 10^{10}$).

Dataset	Method	R	$\rho_{\max}$	κ_2
Lambda	CPG	23	1.10×10^{-2}	5.43×10^{10}
	mCPG	15	8.90×10^{-3}	1.10×10^{3}
	ADG	23	1.10×10^{-2}	5.43×10^{10}
Physics	CPG	9	8.14×10^{-1}	2.10×10^{2}
	mCPG	9	8.06×10^{-1}	2.18×10^{2}
	ADG	11	3.53×10^{-1}	1.39×10^{3}

Two weaknesses. (i) **Accuracy on wide-magnitude data.** On physics, CPG/mCPG rank candidates by *absolute* residual, so they chase a few high-magnitude snapshots and leave the small-magnitude ones badly resolved ($\rho_{\max} \approx 0.8$); ADG's angle criterion already does better (0.35), but not enough. (ii) **Conditioning.** On lambda the snapshots are nearly collinear, so raw-snapshot cones (CPG, ADG) accumulate near-parallel generators and become singular ($\kappa_2 \approx 5 \times 10^{10}$). Only the first of these is an ADG problem with an ADG fix, and Sec. 3 gives it. The second is structural: any method that stores raw snapshots as generators inherits it, so it is answered by mCPG's shifted, re-normalised generators rather than by anything ADG can do to its selection rule.

3 Idea 1 — per-snapshot normalisation

ADG's *selection* is scale-invariant, but its *stopping* rule still compares the absolute residual to a single global scale $\varepsilon \max_x \|x\|$. On wide-magnitude data this lets a small snapshot leave the "unresolved" set while still 50%+ wrong in relative terms. Measuring the error relative to each snapshot's own norm (dividing x by $\|x\|$ before the stopping check, while still storing physical-scale generators) makes the criterion a genuine per-snapshot relative one. It is applied to ADG *only*; CPG/mCPG have no such option and stay on the raw snapshots.

Figure 1 shows the effect on physics: the raw methods leave a broad plateau of small-magnitude snapshots stuck near 10^{-2} relative error, while normalised ADG drives them to machine precision. At $\varepsilon = 10^{-2}$ normalised ADG reaches $\rho_{\max} = 0.143$ (mean 3.7×10^{-3}) versus 0.35 for raw ADG and 0.81 for CPG/mCPG.

On lambda (magnitude ratio 1.13) it is *not* free, and calling it "inert" there — as an earlier version of this note did — understates the cost: it inflates R from 23 to 26 and κ_2 from 5.43×10^{10} to 1.38×10^{11}, buying a $\rho_{\max}$ of 9.52×10^{-3} against 1.10×10^{-2} — a 13% accuracy gain for 13% more generators and 2.5× worse conditioning. That is the same pattern the contact-force sets show (Sec. 5), so it is not a quirk of those datasets: whenever the magnitude ratio is near 1 there is no masked small-magnitude snapshot to rescue, and the stricter criterion only demands more generators. Normalisation earns its keep on physics (ratio 604), not here.

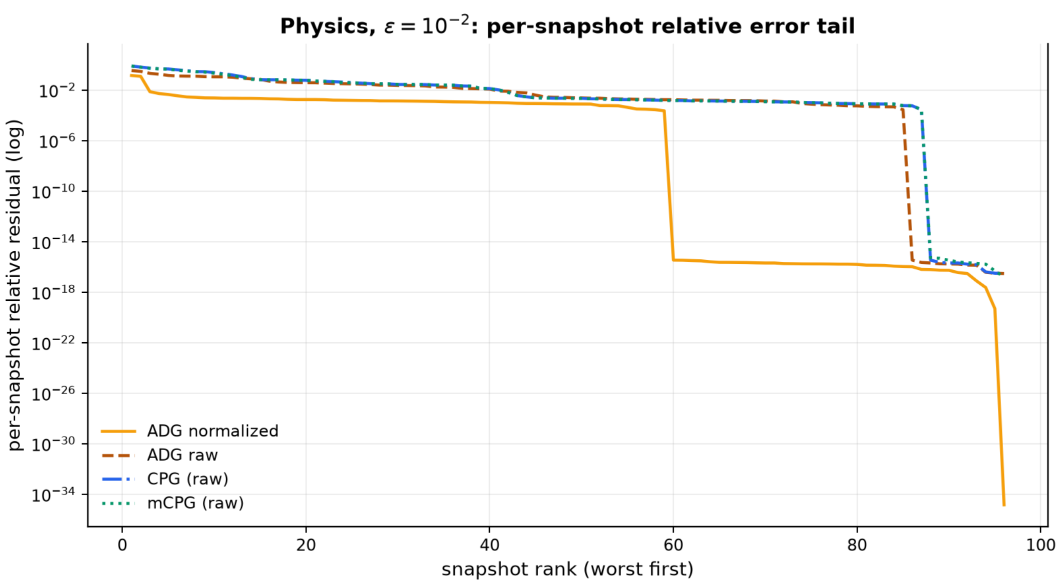

Figure 1: Physics, $\varepsilon = 10^{-2}$: per-snapshot relative residual sorted worst-first (log scale). Normalisation (applied to ADG only) resolves the small-magnitude snapshots that a global-scale criterion neglects, which is what collapses $\rho_{\max}$.

4 The standard ADG and a fair comparison

We define the **standard ADG** as plain ADG with per-snapshot normalisation on. Normalisation is its only setting: it moves the stopping test, never the argmax, so it cannot change *which* generator a round selects — only how many rounds are run before every snapshot counts as resolved. Since the three methods otherwise stop at different R, comparing them at their own stopping points measures the tolerance as much as the method. We therefore walk each basis prefix $K_1, \dots, K_R$ and compare at *matched* cone size (Table 3); all three greedies are nested, so a prefix of length n is exactly the cone that method would have produced had ε stopped it at n.

Table 3: Matched cone size: worst-case relative residual $\rho_{\max}$ and condition number κ_2 for the standard ADG and for CPG/MCPG at the same R (whole dataset). Physics is the wide-magnitude set (ratio 604) and lambda the well-scaled one (ratio 1.13); the two behave in opposite ways. Source: `physics_component_sweep_metrics.csv`, `component_sweep_metrics.csv`.

		ADG		CPG		MCPG	
Dataset	R	$\rho_{\max}$	κ_2	$\rho_{\max}$	κ_2	$\rho_{\max}$	κ_2
Physics	5	1.66×10^{-1}	1.30×10^{3}	8.43×10^{-1}	1.00×10^{2}	8.43×10^{-1}	9.60×10^{1}
	12	5.06×10^{-2}	1.60×10^{3}	7.63×10^{-1}	3.30×10^{2}	7.63×10^{-1}	2.50×10^{2}
	24	2.43×10^{-2}	2.20×10^{3}	5.20×10^{-1}	1.20×10^{3}	5.20×10^{-1}	3.90×10^{2}
	30	1.87×10^{-2}	2.30×10^{3}	5.20×10^{-1}	2.50×10^{3}	5.20×10^{-1}	3.70×10^{3}
Lambda	7	7.98×10^{-2}	1.50×10^{2}	7.98×10^{-2}	1.50×10^{2}	6.60×10^{-2}	1.60×10^{1}
	15	6.75×10^{-2}	4.00×10^{4}	6.75×10^{-2}	4.00×10^{4}	4.91×10^{-2}	4.60×10^{2}
	21	6.75×10^{-2}	4.60×10^{10}	6.75×10^{-2}	4.00×10^{10}	9.62×10^{-3}	3.00×10^{10}
	30	6.75×10^{-2}	4.80×10^{11}	6.75×10^{-2}	4.80×10^{11}	6.55×10^{-3}	7.10×10^{10}

Reading the comparison. On **physics** the standard ADG dominates on accuracy at every budget: its worst-case relative error is 15–28× lower than CPG/MCPG (1.87×10^{-2} versus 5.20×10^{-1} at $R = 30$), because only ADG, through normalisation, attends to the small-magnitude snapshots the other two leave 50% wrong. Note what it does *not* buy: ADG's cone is no better conditioned than CPG's, and at small R it is an order of magnitude *worse* (1.3×10^3 against 1.0×10^2 at $R = 5$). Accuracy on wide-magnitude data is the whole of the claim.

On **lambda** the standard ADG and CPG are indistinguishable — same $\rho_{\max}$ at every R, same κ_2 to within rounding — because with a magnitude ratio of 1.13 there is no masked snapshot for the angle criterion to rescue, and the two selection rules agree. Both become numerically singular past $R \approx 21$. MCPG is the only method that is both more accurate (6.55×10^{-3} at $R = 30$, an order of magnitude better) and better conditioned. On well-scaled data ADG has nothing to add over CPG, and the right choice is MCPG.

5 Confirmation on four contact-force datasets

The three ideas above were developed on *lambda* and *physics*. We now test them on four independent datasets — the membrane-obstacle and Hertz-contact test cases of Niakh, Drouet, Ehrlacher & Ern (ESAIM:M2AN 2022), plus two further reconstructions of the Hertz physics, cases (a) and (b) — where $\lambda(\mu) \geq 0$ *is* the physical contact force. Sections 5 –5.4 report the first two, on which the analysis was done; Sec. 5.5 adds cases (a) and (b). Together they confirm the note's central thesis and force *three* refinements — the third, from case (a), retracting a claim the other three datasets had made look like a law.

What is different here. Three things depart from Secs. 1–5, and each matters for reading the numbers.

- **A non-Euclidean dual inner product.** Errors, projections and angles are measured in $\|x\|_W = \sqrt{x^\top W x}$ with W the dual mass matrix shipped with each dataset, not in $\|\cdot\|_2$. We factor $W = U^\top U$ and run the *unmodified* Euclidean algorithms on U-transformed snapshots: $\|x\|_W = \|Ux\|_2$ and $U(Ac) = (UA)c$ with the same $c \geq 0$, so the transformed runs *are* the W-norm greedies. The lone exception is mCPG's cone-shift constraint $\Upsilon \leq \theta$, which is elementwise and *not* preserved by U; mCPG is given U^{-1} so that bound is imposed in physical coordinates.
- **Held-out testing.** Unlike Tables 2 and 3, which score the whole dataset, these runs fit on a training set and report a held-out $\rho_{\max}$ at unseen μ. Hertz holds out interior R_2 values (65/16); the membrane holds out the strict interior of its $5 \times 5 \times 5$ grid (98/27), since its μ_1 takes only 5 distinct values and a split on it would be degenerate. This turns out to change the conclusion (Sec. 5.4).
- **Feasible tolerances.** $\varepsilon = 10^{-2}$ is unreachable on the membrane: POD needs $R = 97/98$ to resolve the training set at all, so every method simply exhausts it ($R = P$). The membrane is therefore run at $\varepsilon = 0.25$. Hertz uses $\varepsilon = 10^{-2}$ as before.

5.1 Baseline: the lambda pattern, reproduced exactly

Table 4: Baseline on the contact-force sets (ADG raw; Hertz $\varepsilon = 10^{-2}$, membrane $\varepsilon = 0.25$; ρ held out at unseen μ). Hertz reproduces the lambda failure mode exactly: CPG/ADG are numerically singular ($\kappa_2 \sim 10^9$) while mCPG is compact and well-conditioned.

Dataset	Method	R	$\rho_{\max}$ (train)	$\rho_{\max}$ (test)	κ_2
Hertz	CPG	38	1.11×10^{-2}	6.28×10^{-2}	5.51×10^{9}
	mCPG	25	1.13×10^{-2}	4.82×10^{-2}	9.79×10^{2}
	ADG	38	1.11×10^{-2}	6.28×10^{-2}	5.51×10^{9}
Membrane	CPG	56	2.98×10^{-1}	3.56×10^{-1}	4.91×10^{1}
	mCPG	52	2.71×10^{-1}	3.67×10^{-1}	3.07×10^{1}
	ADG	56	2.98×10^{-1}	3.56×10^{-1}	4.91×10^{1}

Table 4 is Table 2's lambda block again, on new data: CPG and ADG are *bit-identical* (they select the same set, in a different order, hence the same cone) and near-singular at $\kappa_2 = 5.5 \times 10^9$; mCPG reaches the same tolerance with 34% fewer generators and a κ_2 seven orders of magnitude smaller. Both datasets have magnitude ratio ≈ 1.2, and both behave exactly as the ratio predicts. This is the note's main claim confirmed on data it was not developed on.

5.2 Idea 1 on new data: normalisation is not merely inert — it is harmful

Sec. 3 found normalisation "inert" on uniform-magnitude data. On these sets it is worse than inert: at ratios of 1.16/1.30 there is no masked small-magnitude snapshot for it to rescue, so the stricter per-snapshot criterion only demands more generators for no return. On Hertz it takes ADG from $R = 38$ to $R = 42$ and κ_2 from 5.5×10^9 to 4.6×10^{11}; on the membrane from $R = 56$ to $R = 82$. Held-out error is unchanged to three digits in both (`adg_variants.csv`). Normalisation here costs 11% more generators and two orders of magnitude of conditioning, and returns nothing.

Refinement. Normalisation should be read as conditional on the magnitude ratio, not as free: switch it on for wide-magnitude data (physics, ratio 604), off otherwise. Since the standard ADG is normalisation, this says the standard ADG is itself conditional — it is the right method only in the wide-magnitude regime.

5.3 Matched cone size

Because all three greedies are nested, the prefix $K_1 \subset \cdots \subset K_R$ is exactly the cone each would return at a looser ε. Walking the prefixes therefore separates "better cone" from "stopped sooner" (Table 5, Fig. 2).

Table 5: Matched cone size on Hertz ($\varepsilon = 10^{-2}$). At equal R, mCPG dominates on conditioning and on both error columns. CPG and ADG are identical at every R *on this dataset* — they select the same set in a different order — but that coincidence is not general (Sec. 5.5). Source: `matched_r.csv`.

R	Method	$\rho_{\max}$ (train)	$\rho_{\max}$ (test)	κ_2
5	CPG	7.17×10^{-2}	6.94×10^{-2}	2.88×10^{1}
	mCPG	7.17×10^{-2}	6.94×10^{-2}	1.45×10^{1}
	ADG	7.17×10^{-2}	6.94×10^{-2}	2.88×10^{1}
12	CPG	4.38×10^{-2}	6.28×10^{-2}	3.40×10^{2}
	mCPG	4.38×10^{-2}	6.10×10^{-2}	6.27×10^{1}
	ADG	4.38×10^{-2}	6.28×10^{-2}	3.40×10^{2}
16	CPG	3.50×10^{-2}	6.28×10^{-2}	4.17×10^{2}
	mCPG	2.94×10^{-2}	6.04×10^{-2}	1.08×10^{2}
	ADG	3.50×10^{-2}	6.28×10^{-2}	4.17×10^{2}
25	CPG	2.11×10^{-2}	6.28×10^{-2}	4.94×10^{3}
	mCPG	1.13×10^{-2}	4.82×10^{-2}	9.79×10^{2}
	ADG	2.11×10^{-2}	6.28×10^{-2}	4.94×10^{3}

At matched R the standard ADG buys *nothing* here: it is CPG, to every digit in the table, because at a magnitude ratio of 1.16 its normalisation has no masked snapshot to rescue and its angle criterion picks CPG's set. The membrane agrees (at $R = 52$: mCPG $\kappa_2 = 3.07 \times 10^{1}$, $\rho_{\max}^{\text{test}} = 3.673 \times 10^{-1}$; ADG 4.25×10^{1}, 3.686×10^{-1}; CPG worst on test at 3.786×10^{-1}). On well-scaled data mCPG is the method of choice at every budget; ADG has no role.

5.4 What only held-out testing reveals: a generalisation floor

Every table in Secs. 1–4 scores the training data. On Hertz that would have hidden the most important effect on these datasets: **the held-out error floors**. CPG/ADG's test $\rho_{\max}$ is pinned at 6.280×10^{-2} from $R = 12$ all the way to $R = 38$ — the last 26 generators improve training error $4\times$ and generalisation by *nothing*. mCPG is the only method that keeps improving out of sample (6.10×10^{-2} at R=12, 6.04×10^{-2} at R=16, 4.82×10^{-2} at R=25).

The floor is the practical result of this section. It says the useful budget on Hertz is $R \approx 12$, and that everything past it is training-set decoration: ε is being asked for an accuracy the data cannot generalise to. It also explains why a smaller cone can look free on this dataset — past the floor, generators cost nothing observable to discard, so any procedure that stops early appears to lose nothing while in fact converging to a training error it never reaches out of sample.

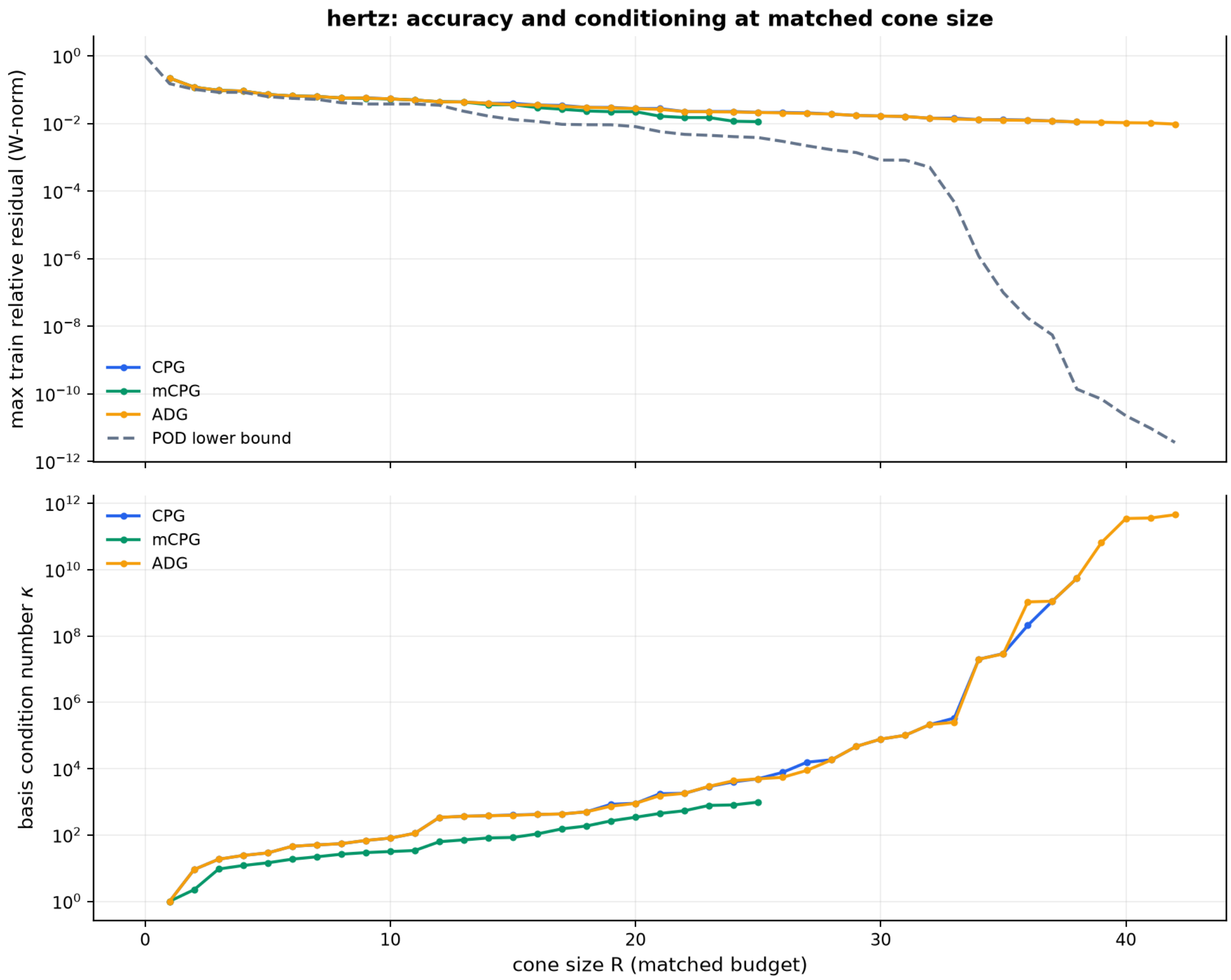

Figure 2: Hertz: accuracy (top) and conditioning (bottom) at matched cone size. mCPG (green) holds a $\sim 4\times$ conditioning margin at every R and stays near 10^3, while CPG/ADG (blue/orange, exactly overlapping) blow up past $R \approx 32$ to 10^{10}.

Why the membrane is hard. Even a cone built from *all* 98 training snapshots leaves a held-out $\rho_{\max} \approx 0.355$, against ≈ 0.12 for an unconstrained POD subspace of the same data. Nonnegative combinations can only add mass, never cancel it, so a reaction bump at an unseen (μ_2, μ_3) is unreachable from bumps centred elsewhere. This is the moving-support regime, and no knob studied here addresses it.

5.5 Two further Hertz cases: where CPG $\equiv$ ADG breaks

Two additional reconstructions of the Hertz physics sharpen the picture. Both put λ on the same reference arc (51 uniform nodes on $[-1, 1]$) but drive it differently: **case (a)** sweeps the *imposed displacement* $\mu = d \in [0.15, 0.45]$ (31 snapshots, held out 25/6), **case (b)** the *geometry* $\mu = R_2 \in [0.905, 1.125]$ (23 snapshots, 18/5). Pressing harder in (a) both scales the semi-ellipse up and widens its support, which is why (a) is the only set here with a non-trivial magnitude ratio (3.93, Table 1); (b)'s narrow range about $R_2 \approx 1$ barely moves the profile (ratio 1.09).

A caveat on the metric. Unlike the other two sets, neither ships a W matrix. Both live on a 1-D contact arc, so W is rebuilt as the P_1 arc mass matrix of the node abscissas — the same construction that produced the Hertz set's own W. Without it the "W-norm" would silently degrade to a mesh-dependent nodal norm, so every number below rests on that reconstruction.

They do not discriminate on accuracy. Both are far lower-rank than Hertz (numerical rank 14 and 8; POD reaches 1.8×10^{-6} by $R = 4$ on (b)). At $\varepsilon = 10^{-2}$ all three methods hit the tolerance at $R = 7$ (a) / $R = 3$ (b) with *identical* errors, so Table 6 reports the discriminating $\varepsilon = 10^{-3}$. Conditioning is the only axis that separates the methods — and mCPG wins it by one to two orders, as on Hertz. These two sets are best read as a conditioning benchmark, not as evidence about accuracy.

Table 6: Hertz cases (a) and (b) at $\varepsilon = 10^{-3}$ (ρ held out at unseen μ). Accuracy ties across methods; only κ_2 separates them, and mCPG wins it everywhere. Case (b)'s mCPG cone is very nearly orthogonal.

Dataset	Method	R	$\rho_{\max}$ (train)	$\rho_{\max}$ (test)	κ_2
Hertz (a)	CPG	8	1.25×10^{-3}	1.44×10^{-2}	1.84×10^{3}
	mCPG	8	9.49×10^{-4}	1.44×10^{-2}	3.71×10^{1}
	ADG	8	1.25×10^{-3}	1.44×10^{-2}	1.84×10^{3}
Hertz (b)	CPG	3	3.06×10^{-4}	2.97×10^{-4}	1.78×10^{2}
	mCPG	3	3.06×10^{-4}	2.97×10^{-4}	2.40
	ADG	3	3.06×10^{-4}	2.97×10^{-4}	1.78×10^{2}

Refinement 3: CPG and ADG are not the same algorithm. Table 4 and Table 5 record CPG and ADG as bit-identical, and lambda said the same; it is tempting to read that as a law. It is not. On case (a) at $\varepsilon = 10^{-3}$ they again coincide — both select $\{0, 3, 6, 10, 15, 20, 25, 30\}$, differing only in order — but at $\varepsilon = 0.05$ they *diverge, and* ADG *loses*:

Method	Selections ($R = 4$)	$\rho_{\max}$ (train)	$\rho_{\max}$ (test)	κ_2
CPG	$\{0, 10, 20, 30\}$	7.36×10^{-2}	3.93×10^{-2}	3.16×10^{1}
mCPG	$\{0, 10, 20, 30\}$	7.36×10^{-2}	3.89×10^{-2}	1.41×10^{1}
ADG	$\{0, 15, 20, 30\}$	1.03×10^{-1}	8.26×10^{-2}	3.76×10^{1}

The angular criterion prefers node 15 over node 10 and that is the wrong call: 2.1× worse held-out error at the same budget, and worse conditioning too. So the identity is a property of these particular datasets, not of the two algorithms — it holds while ε is tight enough that both are forced onto essentially the whole training set, and breaks as soon as ε is loose enough for the choice to bite. Where they differ, plain ADG is the one that loses. Across all six datasets and every ε tested, plain ADG never once beats CPG: it ties or it loses.

Normalisation: a lower bound on where it pays. Case (a)'s ratio of 3.93 is the first intermediate sample between the well-scaled sets (1.1–1.3) and physics (604). Normalisation is still *harmful* there: R goes $8 \to 9$ and κ_2 $1.8 \times 10^3 \to 1.6 \times 10^5$ at unchanged held-out error. A ratio near 4 is therefore not enough to make the per-snapshot criterion pay; the threshold lies well above it, closer to physics' regime than to these.

The floor reappears. Case (a) reproduces Sec. 5.4's effect: held-out $\rho_{\max}$ is pinned at 1.438×10^{-2} from $R = 7$ onward — for CPG and mCPG alike, and for every ADG variant that runs as far as $R = 11$ — while training error keeps falling. That the floor is common to all methods and variants on a second, independently generated dataset supports reading it as a property of the interpolation split rather than of any one greedy.

6 Conclusions

- **Baseline.** Raw greedy cones have two failure modes: poor worst-case relative accuracy when snapshot magnitudes vary (CPG/mCPG on physics), and near-singular bases from near-parallel generators (CPG/ADG on lambda).

- **Normalisation** fixes the first for ADG, and defines the **standard ADG**: on physics it cuts the worst-case relative error from 0.52 (CPG/mCPG) to 0.019 at a matched $R = 30$, a 28× margin that holds at every budget.

- **Nothing in ADG fixes the second.** The conditioning failure is structural — any cone that stores raw snapshots as generators inherits it — so it is mCPG's shifted, re-normalised generators that answer it, not ADG. On lambda the standard ADG is numerically singular at large R exactly like CPG ($\kappa_2 \approx 5 \times 10^{11}$ at $R = 30$), while mCPG is both an order of magnitude more accurate and better conditioned.

- **Standard ADG** is therefore a specialist, not a default: best on wide-magnitude data at every matched cone size, and indistinguishable from CPG everywhere else. *Guidance:* use the standard ADG when snapshot magnitudes span orders of magnitude; use mCPG otherwise — on well-scaled data it leads on both accuracy and conditioning, and ADG adds nothing over CPG.

- **Confirmed, with two refinements** (Sec. 5). Four independent contact-force datasets reproduce the lambda pattern exactly — magnitude ratio ≈ 1.2, CPG/ADG identical and singular ($\kappa_2 = 5.5 \times 10^9$), mCPG compact and well-conditioned — so the magnitude ratio predicts the winner on data the method was not tuned on. But: (i) normalisation is *harmful* rather than inert when the ratio is near 1 (R: $38 \to 42$, κ_2: $5.5 \times 10^9 \to 4.6 \times 10^{11}$ on Hertz), so the standard ADG is itself conditional on the regime; (ii) **CPG ≡ ADG is not a law** (Sec. 5.5). It holds on lambda, Hertz and Hertz (b), but breaks on Hertz (a) at $\varepsilon = 0.05$, where ADG makes a different — and 2.1× worse — held-out choice at the same $R = 4$. Plain ADG ties or loses to CPG on every dataset tested; it never wins.

- **Where normalisation starts to pay.** Hertz (a) is the only sample at an intermediate magnitude ratio (3.93), and normalisation is still harmful there (κ_2: $1.8 \times 10^3 \to 1.6 \times 10^5$). The ratio at which the per-snapshot criterion begins to earn its keep is thus bounded well above 4 — the physics set (604) remains the only evidence that it ever does.

- **Train-only metrics hide the main effect.** With held-out parameters, CPG/ADG's test error is flat from $R = 12$ to $R = 38$ on Hertz: 26 generators improve training error 4× and generalisation not at all. Only mCPG keeps improving out of sample. Any future knob should be judged against held-out error at matched R, not training error at its own stopping point.

Reproducibility. Matched-R comparison (Table 3): `results/physics/reduction/physics_component_sweep` and `results/lambda/component_sweep/component_sweep_metrics.csv`, from `python -m greedy.pipelines.ph` and `python -m greedy.pipelines.component_sweep`. Stopping-point runs: `results/{lambda,physics}/reduct` and `results/lambda/adg/`. Normalisation study (Fig. 1): `results/physics/adg_normalization/` and `results/lambda/adg_normalization/`. All numbers are taken verbatim from the `*.csv` and report files there.

Two caveats a reader re-running this should know. **(i)** The script that produced the two `adg_normalization/` directories was never committed, so they cannot be regenerated from the repository; Fig. 1 rests on them. The effect they illustrate is independently reproducible from `physics_component_sweep_metrics.csv`, which is generated by a committed pipeline and carries the same conclusion at matched R. **(ii)** `results/synthetic/` was, until this revision, irreproducible: the snapshot generator drew from the unseeded global `np.random`, so every run produced different data while `pipelines.synthetic`'s `rng_seed` only fixed the train/test split. The generator now takes an explicit seed and the directory is deterministic; its numbers therefore differ from any pre-revision copy, and no claim in this note depends on them.

Contact-force sets (Sec. 5): `results/contact_forces/` (`comparison_metrics.csv`, `adg_variants.csv`, `matched_r.csv`, `matched_r.png`, `convergence_vs_pod.png`), regenerated with

```
python -m greedy.pipelines.contact_forces_compare --case hertz \
       --epsilon 0.01 --adg-study --matched-r
python -m greedy.pipelines.contact_forces_compare --case membrane \
       --epsilon 0.25 --adg-study --matched-r
```

Hertz cases (a) and (b) (Sec. 5.5): `results/contact_forces/hertz_{a,b}_eps{0.05,0.01,0.001}/`, regenerated with

```
for case in hertz_a hertz_b; do
  for eps in 0.05 0.01 0.001; do
    python -m greedy.pipelines.contact_forces_compare --case $case \
           --epsilon $eps --adg-study --matched-r \
           --output-dir results/contact_forces/${case}_eps${eps}
```

```
  done
done
```

Dataset provenance and the W-norm integration are documented in `docs/contact_force_datasets.md`; the correctness properties the transform relies on are pinned in `tests/test_contact_forces.py`, including the P_1 arc-mass reconstruction of W that cases (a) and (b) depend on.

A note on the timing columns. The `elapsed_seconds` column of `comparison_metrics.csv` was long unusable: the three methods are fitted in a fixed order, and the first one run absorbed the cost of spawning a `joblib` worker pool (≈ 1.6 s), which made CPG look uniformly slower than it is. The parallelism turned out to be a pessimisation on every workload here — each per-row NNLS solve takes microseconds while joblib pickles the basis matrix per row — so it was removed; serial is 5–260× faster with bit-identical results. The column is now a fair comparison, and it says something the old numbers hid: mCPG is the *slowest* of the three (membrane, $\varepsilon = 0.25$: CPG 0.68 s, ADG 0.18 s, mCPG 5.51 s), because its constrained cone-shift QP runs once per generator. That cost is the price of the conditioning win in Table 4.